



المهية العربية للتصنيع

ARAB ORGANIZATION
FOR INDUSTRIALIZATION



الهيئة العربية للتصنيع

ARAB ORGANIZATION
FOR INDUSTRIALIZATION

AOI Business Fields (Defense Industries)

١ General

٢ AOI Activities Development

٣ AOI Vision

٤ AOI Collaboration with Private Sector (Egyptian- Arab - Foreign)

٥ AOI Technological and Manufacturing Capabilities in Defence Industries

٦ AOI Facilities to Qualify the Human Cadres

٧ AOI Technological and Manufacturing Capabilities in Civil Industries

General

Arab Organization for Industrialization (AOI)

An International Arab Organization
with special regulations

Established

1975

**Egypt - Saudi Arabia - Qatar -
United Arab Emirates**

Co-founders

**Localize Advanced Arab
Defense Industries**



AOI has become
**an Egyptian State-Owned
Organization**
under the same name,
laws and regulations

**In May
1994**



AOI invites sisterly and friendly
countries to join the organization

AOI Supreme Committee

Chaired by **H.E. the President of Egypt**, and includes the following members: the **PM** and Cabinet Ministers

- ❖ Deputy Prime Minister for Industrial Development & Minister of Transport and Industry
- ❖ Commander-in-Chief of the Armed Forces & Minister of Defense and Military Production
- ❖ Minister of Planning, Economic Development, and International Cooperation
- ❖ Minister of Foreign Affairs and Immigration
- ❖ Minister of Finance
- ❖ Minister of State for Military Production
- ❖ Chief of Staff of the Armed Forces



The Supreme Committee sets AOI strategy

AOI Board of Directors

☐ **Headed by the Chairman of AOI** and includes the following members :

- ❖ Governor of the Central Bank of Egypt
- ❖ Dr. Mahmoud Esmat – Minister of Electricity and Renewable Energy
- ❖ Chief of Armed Forces Operations Authority
- ❖ Chief of Armed Forces Armament Authority
- ❖ Deputy Minister of Finance for Tax Policy
- ❖ Chief of Industrial Development Authority
- ❖ Chairman of Egypt Air Holding Company
- ❖ Chief of Air Force Aviation Engineering Department

- ☐ **AOI General Manager** presides over the Secretariat of the Council
- ☐ The BoD implements the strategy and policies of the Supreme Committee

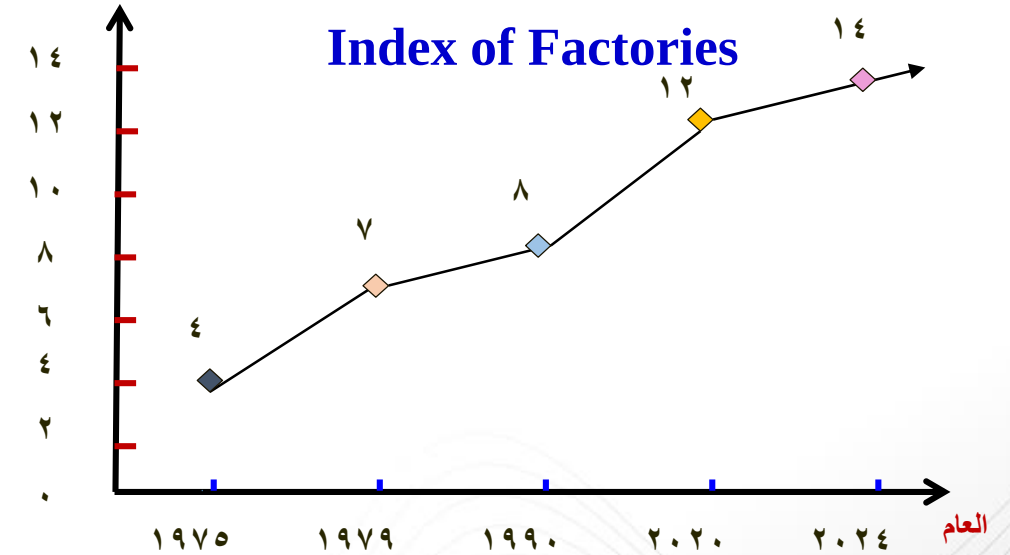
AOI Activities Development

AOI Activities Development

❑ AOI is the largest industrial hub and one of the national security pillars of Egyptian economy

❑ AOI has been growing to include (14) factories and companies

No. of factories



Defense fields



Civil fields



**(7) factories/
companies**
operating
exclusively in
civil
industries

AOI owns
(14)
**factories/
companies**

**(7) factories/
companies**
operating in
defense and
civil
industries

Cont. AOI Activities Development

Sakr Factory for Advanced Industries

Kader Factory for Advanced Industries

Aircraft Factory

Helwan Factory for Advanced Industries

**Digital Manufacturing
Center (DMC)**

Engine Factory

Electronics Factory

Arab British for Dynamics Industries (**ABD**)

**Factories
Operating in
Defense and
Civil
Industries**

SEMAF Factory for Railway Equipment

ATICO Factory for Timber Industries

Arab Company for **Real Estate and Tourism Projects**

Arab **Renewable Energy** Company

Arab **African Company for Medical Industries**
the root of a medical industrial city
(Under Construction)

Arab **American Vehicles** Company (**AAV**)

Arab **German** Company for Secured License Plates (**AGC**)

**Factories
operating
exclusively
in civil
industries**



An Organization Aiming to Modernization & Technology Development
to Achieve the Goals of the New Republic

AOI Vision in Collaboration with partners

AOI Liability

Provides (land, utilities, skilled labor), and building constructions.

Partner Liability

Provides (know-how) - equipment – labour - raw materials - ensures quality control - technical and marketing management.

Key Partnership Parameters

The partner's liability is to export portion of the production to (procure imported raw materials – allocate foreign partner's profit)



AOI Collaboration with (Egyptian – Arab – Foreign) Private Sector

Egyptian Private Sector





**Awazel Company
(KSA)**

**Advanced Nano
Technology** Products for
Protecting Various
Surfaces



مجموعة طلال أبوغزاله
Talal Abu-Ghazaleh Organization

**Talal Abu-Ghazaleh
Organization
(Jordan)**

**Laptops, Tablets, and
CellPhones**



**ZEED Company
(Oman)**

Defense Products

Foreign Private Sector

	 北方工业 NORINCO  中國保利集團有限公司 CHINA POLY GROUP CORPORATION LIMITED	 DASSAULT AVIATION  RUSSIAN HELICOPTERS  SAFRAN	Defense Industries
Armored Vehicles	Missiles	Aircraft	

				Civil Industries
Railways and Underground Metros				
				
License Plates	Wood Working	Machinery	Elevators	

AOI Technological and Manufacturing Capabilities in **Defense Industries**

AOI Technological and Manufacturing Capabilities in Defense Industries

All Defense Products are **Exclusive Intellectual Property to the Arab Organization for Industrialization (AOI)**



Cont. AOI Technological and Manufacturing Capabilities in **Defense Industries**

Aircraft

Engine Overhaul

Armored Vehicles

Ammunition and Missiles

Electronic Equipment

**Defense
Industries**

Aircraft Industry / Helicopters

Cont. AOI Technological and Manufacturing Capabilities in Defense Industries



Tucano

1983-1990



Gazelle Helicopter

1985-1986



Alphajet

1981-1985



Al-Qahira 300

1959-1969



Al-Qahira 200

1959-1969



Al- Gomhoreya

1950



K-8E

2000- present



ASN-209

2000-2009



حلوان ٣

1991

1950-1990

1991-present

Cooperation with the French Company Dassault Aviation



- ☐ parts of Aircraft are manufactured at the AOI factories for the French company Dassault.
- ☐ One of these parts was displayed by Dassault at the Le Bourget Airshow in France.

- ☐ This step qualifies AOI to join the global supply chain system of the aircraft industry.

Overhaul and Maintenance of Aircraft / Helicopters

Cont. AOI Technological and Manufacturing Capabilities in Defense Industries

Overhaul and Maintenance of Helicopters (Eastern/Western)



AOI owns 2 advanced centers for performing the overhaul of helicopters (Eastern-Western) (Mi-8 / Mi-17 / Gazelle), or any other different types

The two centers are accredited by the countries of origin

Engine Overhaul

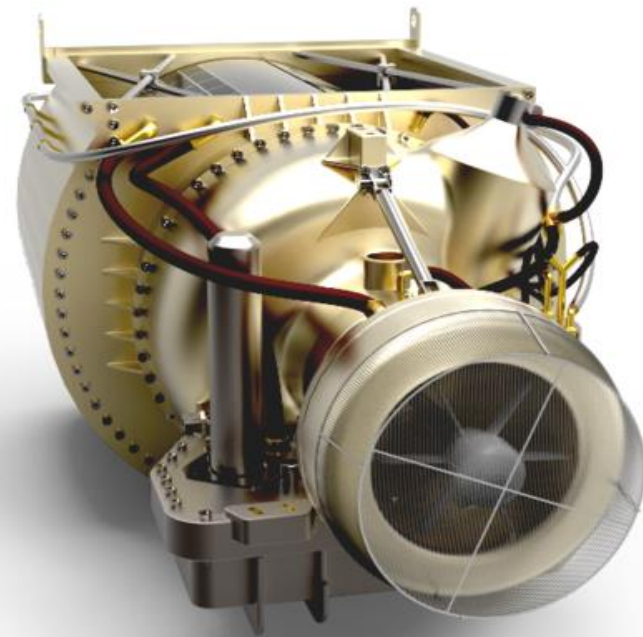
Overhaul / Maintenance of Aircraft Engines



AOI owns (17) overhaul lines of different types of engines, which perform overhaul and maintenance for 90% of the Egyptian Air Force aircraft engines.

All lines are approved by the countries of origin.

Overhaul and Maintenance of Tank Engines



AOI performs overhaul and maintenance for M1A1 tank engines

Armored Vehicles

Armored Vehicles



Designing, producing, and developing the armored vehicle **(Fahd)**, which was exported to many Arab and African countries



Participated with the Egyptian Peacekeeping Forces in **(DRC-Somalia-Sudan-...)**

Cont. AOI Technological and Manufacturing Capabilities in **Defense Industries**

- **Missiles**
- **Launchers & Projectiles**
- **Ammunition**

Missiles

122mm Artillery Missiles



AOI owns
production
lines of **122
mm** Sakr
Missiles
range up to **45
km**

Type	Sakr-45	Sakr-18	Sakr – Thermobaric (thermal pressure)
Range	20-45 km	6-20 km	6-20 km

122mm Smoke Screen Generating Rocketry System



٦٠٠٠/د

- ❑ This Rocketry system generates a smoke screen to protect troops movement and blind enemy observation posts

- ❑ Duration of generating smoke screen (5 – 15) minutes with a range of (4-6) km

Illumination Rocketry System



Illuminates the battlefield at night

Range:(1-5) km

**Duration:
60 Secs**

**Intensity:
million Cd**

Launchers & Projectiles

Cont. AOI Technological and Manufacturing Capabilities in Defense Industries

A light fourfold missile launcher for 122mm artillery missiles launching

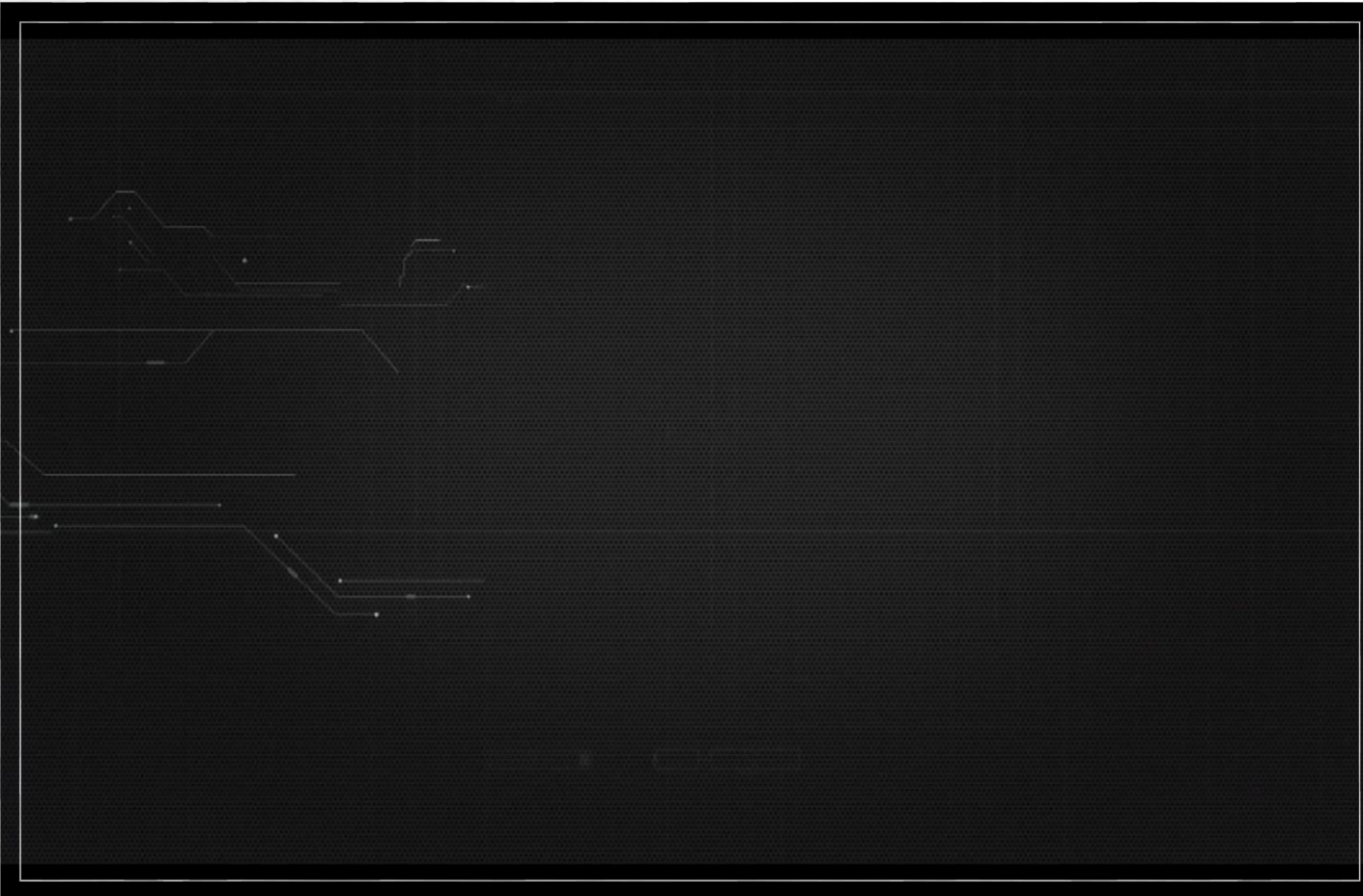
On vehicle



On ground



40 mm RPG-7 Launcher



Over 50 years, the RPG & its projectiles of all types (practice, anti-armored/flammable targets, anti-personnel targets) have been manufactured with a range of (350-1000) m

PG-7 Projectiles

Penetration Depth

Practice

Greater than 26 cm

Greater than 50 cm

Greater than 65 cm

Armored
steel

Anti-flammable targets

Anti-personnel
25-meter effective killing radius

Range

350 m

1000 m

Projectile Type

Practice

Operation

Cobra

Cobra Tandem

Incendiary

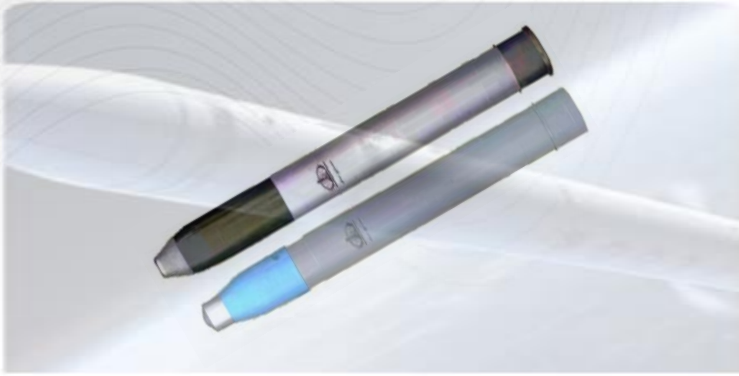
Home Guard

Ammunition

(Military Engineering - Air Force)

Air Force Ammunition

Free Airborne Missiles (Air / Surface)



70 mm Missile

70 mm
Practice /H E

Used with
Apache aircraft

68 mm Missile

68 mm
Practice /H E

Used with
Mirage 5 / Alpha jet aircraft

57 mm Missile

57 mm
Practice /H E

Used with
MI 17-K8 aircraft

Military Engineering Ammunition

Mines and Mine Dispensing Systems



Mine ground dispenser 9



Mine ground dispenser 6



Mine dispenser (2) mounted
on a medium-duty vehicle
with (600) mines



Anti-Tank
Electronic Mine



Mine dispenser (1) mounted
on a medium-duty vehicle
with (100) mines

Cont. AOI Technological and Manufacturing Capabilities in Defense Industries



Opening (2) paths with dimensions
(120m x 8m)

Fateh 3



Opening a path with dimension
(90m x 4m)

Fateh 4

Path Openings for Personnel



Opening a path with dimensions
(120m x 8m)

Fateh 2

Latest Defense Products



Airborne Bombs (Hafez Series)

Unguided bombs weighing up to **(1000 kg)**, with the ability to penetrate reinforced concrete up to **(180 cm)**. Manufactured in collaboration with **the Egyptian Air Force.**

Concrete Wall Penetration Live Demo
1.8 m



Cont. Airborne Bombs (Hafez Series)

Notes	Total Weight (kg)	Type
	250	Hafez 1
	1000	Hafez 2
 	1000 (Bunkers Penetration)	Hafez 3
	Impact Fuse/Electronic Timer	Hafez 4

Multi-tasking Automated Command and Control Post



- ❑ Designing and manufacturing automated command posts equipped with the latest software and technologies (100% Egyptian)

- ❖ In the military field (automated command and control posts - monitoring / controlling the airspace)

- ❖ In the civil field at the level of governorates and the competent ministries to link them to the operations rooms of utilities (water - electricity - gas - ambulance - traffic - .. etc.)



Armored Anti-Riot Control Vehicle (Kader-1)

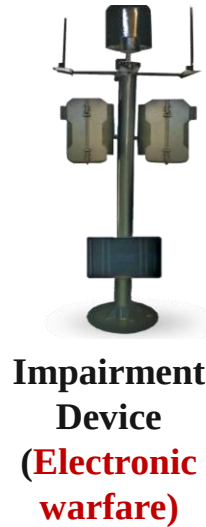


Manufactured in Kader Factory, affiliated to AOI, for the first time in Egypt, with a level of protection (B4) and impedance up to (7.62 x 39).



Extra equipment were added upon the requirements of the Ministry of Interior. It can be customized to meet the client's need.

Anti-Drone and Electronic Impairment System (Hares)



In collaboration with **Egyptian Armed Forces**

Detecting and anti-drones systems up to **the tactical level**

Designing and manufacturing

Anti-drone devices to secure **naval vessels** at different ranges

Manufacturing

Wireless reconnaissance and impairment devices at the different frequency range



Hares



Hares



Hares



Hares

New Egyptian Radiological Detection and Measurement Device



Manufacture as an outcome of fruitful cooperation between AOI & Chemical Warfare Department for the first time in Egypt

❑ **Military Applications**

Up to the sub-unit level and its equivalent

❑ **Civil Applications:**

- ❖ Radiation detection of imports at (border crossings, ports and airports)
- ❖ Hospital radiation therapy departments

Two models of this device have been manufactured

**Key Accreditations and Quality Certificates
Obtained by AOI's Factories
In Defense Industries**

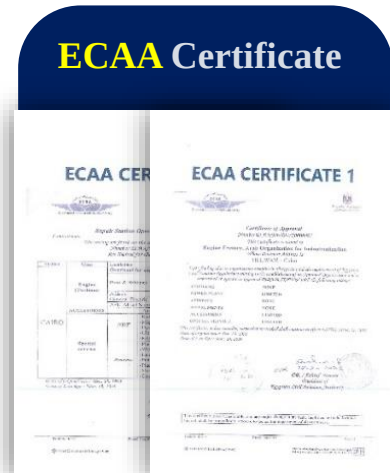
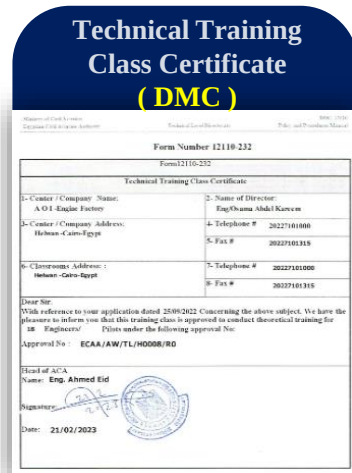
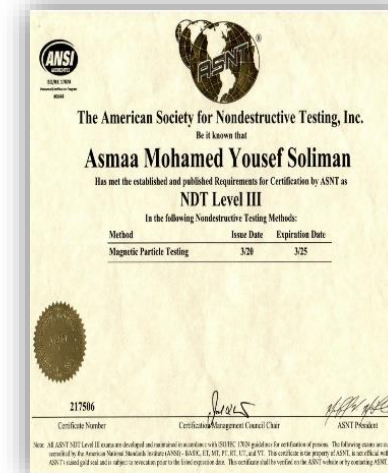
Cont. AOI Technological and Manufacturing Capabilities in Defense Industries

Accreditations and Quality Certificates

 Euro Certificate ISO 9001 - ISO14001 - ISO 45001		Federal Aviation Administration FAA T56-CT64-PT6		Egyptian Civil Aviation Authority ECAA T56-CT64-APU-PT6	
American Welding Society		Egyptian Welding Academy		Institut de Soudre Group	
French Safran Group LARZAC Alphajet aircraft ATAR Mirage-5		English Rolls Royce aircraft T56-A-15LFE (AK: C-130) Gnome H 1400 Hel Commando Aircraft		American General Electric CT64/820/4 Buffalo Plane	
TV3-117VM-S02 Mi-17 Hel.Klimov		Pratt & Whitney PT6A-25C Tucano Aircraft PT6A- 65B Beechcraft Aircraft		American Honey Well APU- GTCP-85-180/L c-130 Aircraft c-130 K8E AGT Aircraft TFE731-2A-2A M1A1 Tanks '1500'	
VO-540-CA S.E Aviation Heller Aircraft AEIO-360-B1B Groub Aircraft O-540-J3A5		Turbomeca Astazou 14H Gazelle Helicopter		Russian Motor Sich R13-300/R25-300 MiG-21 Aircraft	

Cont. AOI Technological and Manufacturing Capabilities in Defense Industries

Cont. Accreditations and Quality Certificates



Cont. AOI Technological and Manufacturing Capabilities in Defense Industries

Cont. Accreditations and Quality Certificates

Overhaul line accreditation certificates for **the Mi-17 Hel.** from the Russian Helicopter Company



Readiness certificates of the factory as an overhaul centre for **the Mi-8 Hel.** from the Russian Helicopter Company



Factory accreditation certificate from **Airbus** to conduct the extended inspection of **Gazelle Hel.**



Factory engineers accreditation and training certificates





الهيئة العربية للتصنيع

ARAB ORGANIZATION
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Civil Industries

Technological and Manufacturing Capabilities

Means of Transport

Electronics

Manufacturing communication towers

Green Energy

Medical Supplies

Pumps Designing and Manufacturing

Pipes Manufacturing

(Water- Sewage- Natural Gas)

Woodworking



**Civil
Industries**

Means of Transport (Railways Coaches and Subways)



- ❑ **Manufacture and assembly of all types of railways coaches (Passengers - Freight wagons - Crops ...etc.), Subways**

- ❑ **Manufacture of railway spare parts (brake - Buffer systems) under license from German company (Knorr-Bremse) and Polish company (Exton)**



- ❑ **Manufacture of high precision large spare parts used in the industry of (ships, turbines, power plants, iron and steel plants, cement factories)**

Cont. Means of Transport (Cars)



- ❑ Cooperation with the two companies (Stellantis / France - Toyota / Japan) to manufacture / assemble private passenger cars (Jeep Cherokee - Toyota Fortuner).
- ❑ New produced models are under study
- ❑ (45%) local component percentage has been achieved



Cont. Means of Transport (Cars)

مصنع قادر للصناعات المتطورة

❑ Equipping all types of vehicles

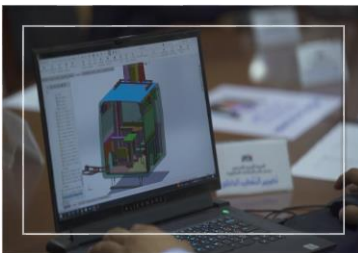
❖ Fire Fighting

❖ Ambulance

❖ Money and documents transfer via armored vehicles

❖ Mobile **ATM** for Banks

❖ Mobile Technological Centers



Cont. Means of Transport (Manufacturing of Secured License Plates)



Production and manufacturing of **Secured License Plates** for all means of transport required by the **Egyptian Ministry of Interior**, in partnership with the **German Company, Tönnjes**



❑ **Contracting the Armed Forces National Service Projects Organization to supply (60) agricultural harvesters for phase 1.**

❑ **(48) harvesters have been delivered**





☐ All Kinds of Security Cameras

☐ Interactive Screens

☐ TV Screens

☐ PCs, Tablets, Cellphones

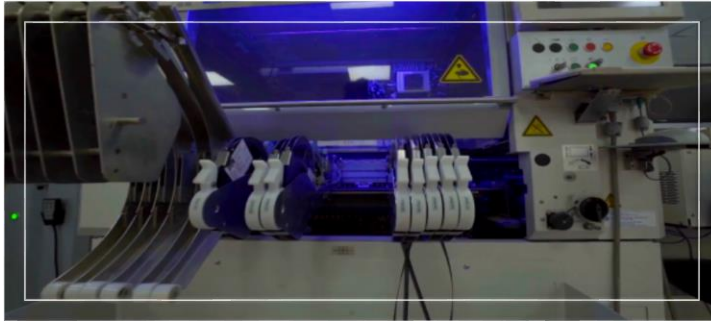
☐ Tact Device for Converting Regular Screens and Surfaces into Interactive Screens.

☐ Manufacture of Electronic Chips for Electrical Devices

☐ Smart Meters (Electricity- Water)

☐ Lightening Systems (LED)

☐ Fiber Optics

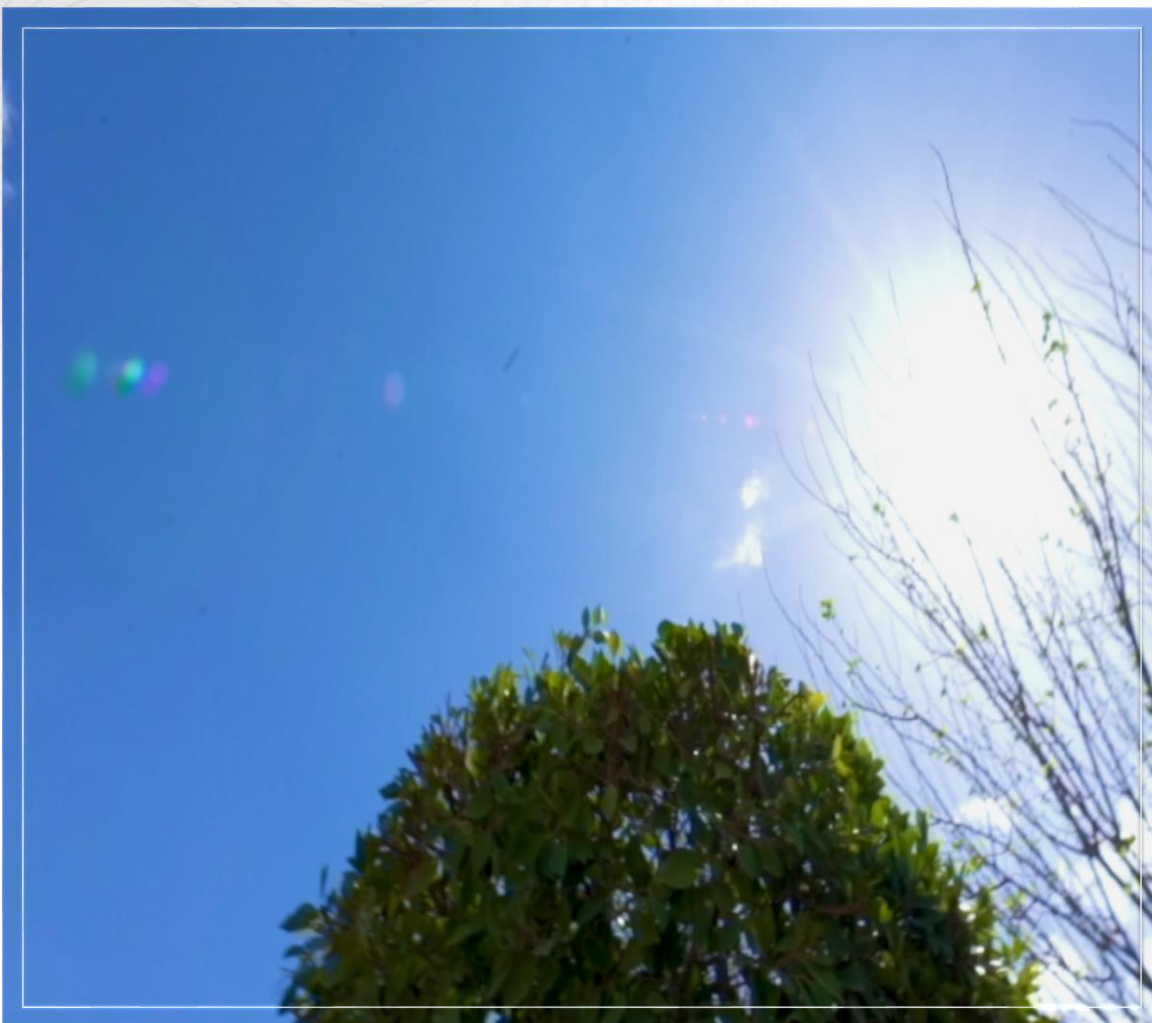


Manufacturing of Cellphone Towers



- ❑ **Cellphone towers manufacturing center** has been inaugurated at the Aircraft Factory
- ❑ **Rooftop** Cellphone towers have been manufactured with **(21) meters high**
- ❑ **Green Field** Cellphone towers are currently produced with **(40 - 80) meters high**
- ❑ **(67)** towers have been delivered for **Telecom Egypt.**





- ❑ Manufacture of solar energy panels. More than (200) plants have been implemented nationwide with a total capacity of nearly (40) megawatts for (Strategic Headquarters - Grand Egyptian Museum)
- ❑ Installation of (84) stations for ambulance centers and (4150) solar-powered lighting poles.
- ❑ In Uganda, a solar power plant has been implemented with a capacity of (4) megawatts
- ❑ In Djibouti, a solar power plant is under contracting with a capacity of (300) megawatts

Safe Gas Valves



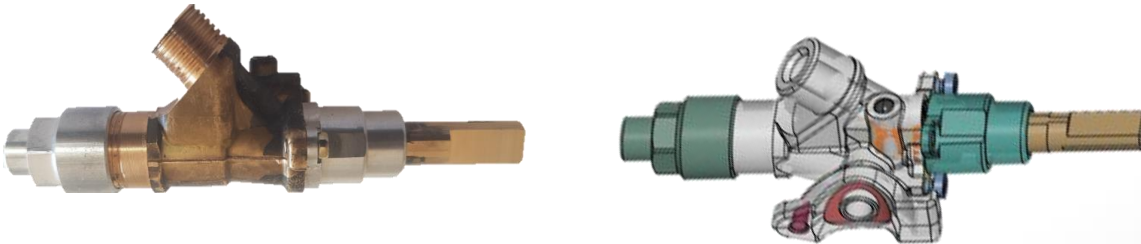
- ❑ A production line for safe gas burners' valves has been inaugurated at the Arab British Dynamic Industries Company

**Production
Capacity**

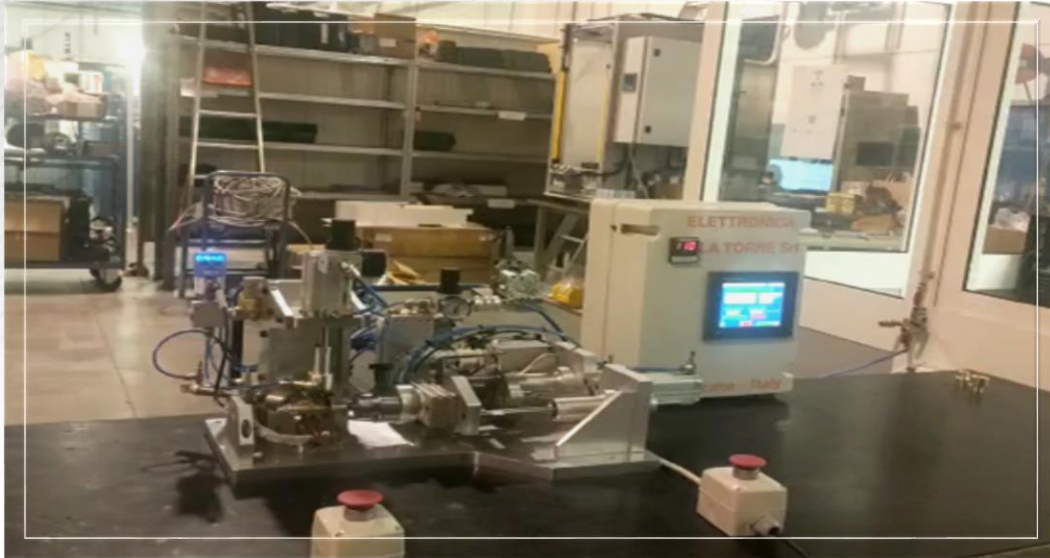
**One million
valves/ year**

Future Plan

**Five million
valves/ year**

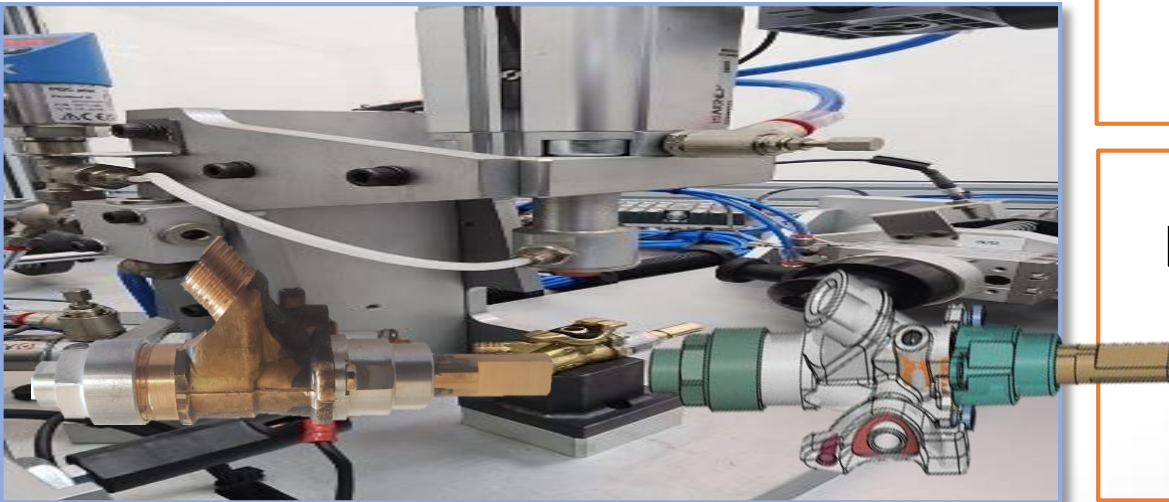


Cont. Manufacturing of Gas Safe Valves



- ❑ Gas safe valves have been tested and accepted by standard **Italian** labs

- ❑ Production of **safe gas valves** to meet the needs of stove manufacturers (Cookers), in coordination with the **Chamber of Engineering Industries / Federation of Egyptian Industries.**



- ❑ Production of valves for **chlorine cylinders** to meet the needs of the local market and for export.

Natural Gas Igniters



Manufacture Natural gas Igniters to operate (steam boilers, melting furnaces, brick factories, foundries, etc.).



A total of **3000 bakeries** have been converted to operate on **natural gas** instead of **diesel and mazut** in collaboration with **the Ministry of Supply and Internal Trade**

Manufacturing of Gas Meters



❑ Manufacture of **household gas meters (prepaid/smart)** that comply with the standard specifications of the Egyptian Natural Gas Holding Company (**EGAS**)

❑ AOI has obtained European certificates of conformity (**EN1359 - EN16314**)

❑ It is planned to produce **(30) thousand meters** during the year **2024/2025** as **an initial phase**

❑ Quantitative production is expected to reach **(2) million meters/ year**

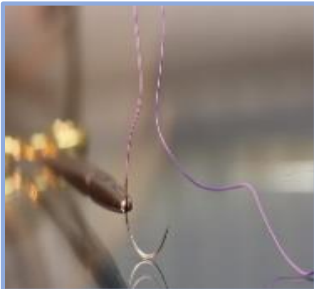


❑ Current Products

- ❖ Oxygen Pumps
- ❖ Medical Beds
- ❖ Incubators
- ❖ Medical Shredders
- ❖ Prefabricated Modular Operation Rooms

❑ plants under Construction

- ❖ Artificial Joints
- ❖ Medical Plates and Screws
- ❖ Smart Syringes
- ❖ Sutures



Designing and Manufacturing **water & Sewage Pumps**



Submersible



Vertical



Horizontal



Split

Local manufacturing
percentage

(78%)

Production capacity

(200 pumps/ month)

Manufacturing of (**Non-Standard**) Spare Parts for Major Companies and Factories

Pump spare parts by reverse engineering

**Mechanical & Electrical
Department**

Spare parts for (Egypt Air - Ezz Steel - Elsewedy Electric - Qous factory – Shotmed company -..)

**Egyptian National
Companies**

Spare parts for (French Dassault - Jordanian Petra Aerospace)

**International/Arab
Companies**

Pipes Manufacturing (Water- Sewage- Natural Gas)



❑ **Manufacturing polyethylene pipes** for water and waste water nets with (23 mm diameter to 1600 mm), with pressure up to (20) bar

❑ **Natural gas nets** from (32 mm and up to 630 mm)

❑ **Manufacturing All Natural gas nets connections**

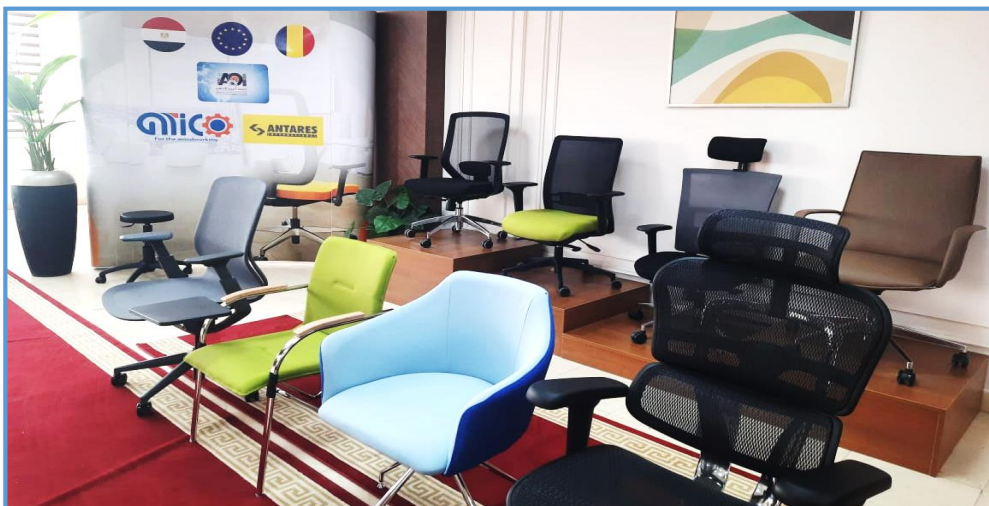
❑ **AOI owns 12 polyethylene pipe production lines** with a production capacity of (103) thousand tons annually.

Wood working Industries



Manufacturing and production of office, hotel, and home furniture

Production lines were inaugurated after developing ATICO Woodworking Company



An agreement was concluded with the Romanian company ANTARES/ to establish a hydraulic chair production line with a local component ratio of up to (75%) to meet the local market needs.



Furnishing (the House of Representatives and Senate), (58) ministries/agencies of the Government District in the New Administrative Capital, in collaboration with distinguished private sector companies.

Significant Projects by Arab Company for Real Estate and Tourism Projects



**Central Labs -
Benha
University**



**Sharm
ElSheikh City
Council**



**Cancer
Hospital-
Tanta
University**



**Administrativ
e Building -
Japanese
University**

AOI Facilities to Qualify **the** **Human Cadres**

AOI Facilities to **Qualify the Human Cadres**

AOI Training Academy

**Arab Institute for Advanced
Technology**

**Technical High School for Dual
Education**

Aero School of Industrial Applied Technologies
in the field of aircraft manufacturing, in
collaboration with **the French company**
Dassault.

**Qualifying
the Human
Cadres Via**



AOI Training Academy

Cont. AOI Facilities to Qualify the Human Cadres



- ❑ Training engineers and technicians on smart manufacturing technologies in collaboration with :

DMG MORI

DMG MORI - Germany

SIEMENS
Ingenuity for life

SIEMENS - Germany



HUAWEI - China

- ❑ The Academy holds an international accreditation certificates



❑ The academy aims to:

Train engineers and technicians on
the work mechanisms of the
Fourth Industrial Revolution
throughout design and digital
manufacturing as well as
operating **CNC machines**



Cont. AOI Facilities to Qualify the Human Cadres



❑ The Academy's contribution to qualification

❖ Training more than **(1500) engineers / technicians annually** (AOI employees - Egyptian factories - engineering faculties students - technical schools students).

❖ Training **engineers** from friendly African countries (**Sudan - South Sudan - Nigeria - Ghana - Namibia - Tanzania - Zambia - Zimbabwe - Cameroon - Mauritius - Sierra Leone**) in the fields of (AI - IT Networks - New and Renewable Energy) **in collaboration with** the Egyptian Agency of Partnership for Development / Egyptian Ministry of Foreign Affairs



Arab Institute for Advanced Technology

Cont. AOI Facilities to Qualify the Human Cadres



❑ The institute aims to:

- ❖ **Qualifying workers** to comprehend and apply mechanisms and procedures of digital transformation
- ❖ **Conducting specialized training courses** to qualify employees in various activities in accordance with new laws and legislation
- ❖ **Qualifying AOI Cadres and Leaders**
- ❖ **Meeting the needs for the workers hierarchy** of various posts, starting from joining AOI and ending with supervisory positions



AOI Technical High School for **Dual Education**

Cont. AOI Facilities to Qualify the Human Cadres



- ❑ In 2019, the school was established in collaboration with Ministry of Education and Technical Education to prepare a generation of qualified and trained technical technicians capable of applying new technologies



- ❑ Theoretical training is conducted in accordance with the curricula of Ministry of Education and Technical Education
- ❑ Practical training is conducted in AOI factories (60% of training hours)



**AOI Aero School of
Industrial Applied
Technologies in
collaboration with
the French
Company Dassault**

Cont. AOI Facilities to Qualify the Human Cadres



Aero School of Industrial Applied Technologies was established in the field of manufacturing aircraft fuselage, starting from the academic year 2023/2024, in cooperation with:

Develop the theoretical curricula and accredit the practical curricula prepared by Dassault

**Ministry of
Education and
Technical
Education**

Provide technological and laboratory capabilities, infrastructure and human resources in accordance with **Dassault Aviation standards**

AOI

Develop specialized curricula in the field of aircraft industry, including : (aircraft fuselage - electrical harnesses - hydraulic system) with full supervision on the educational process for a period of (5) years to qualify the Egyptian trainers

**Dassault
Aviation**



Thank you